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EC39002

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Experiment 5: Quadrature Phase Shift Keying (QPSK) Modulation and Demodulation



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1 Aim of the Experiment

The objectives of this experiment are as follows:

1. Design and implement the circuit for QPSK modulation.
2. Demodulate the modulated signal using coherent detection in MATLAB.

2 Components Used

The following components were used to design the QPSK modulation circuit:

- IC 7404 (Inverter)
- IC M4016 (Switch)
- IC 741 (Op-amp)
- IC 7474 (D flip-flop)
- IC 7486 (Quad 2-input XOR gate)
- IC 74164 (7-bit shift register)
- Resistors and Capacitors

3 Theory

3.1 Phase Shift Keying (PSK)

Phase Shift Keying (PSK) is a digital modulation technique in which the phase of the carrier signal is changed by choosing either the sine or cosine signals at a particular time. This technique is widely used for wireless LANs, biometric devices, contactless operations, RFID, and Bluetooth communications.

3.2 Quadrature Phase Shift Keying (QPSK)

Quadrature Phase Shift Keying (QPSK) is a variation of BPSK and operates as a Double Side Band Suppressed Carrier (DSBSC) modulation scheme. It transmits two bits of digital information at a time, referred to as dibits (bit pairs). Instead of converting digital bits into a simple serial stream, QPSK groups them into bit pairs. This effectively decreases the data bit rate to half, which conserves bandwidth and allows space for other users.

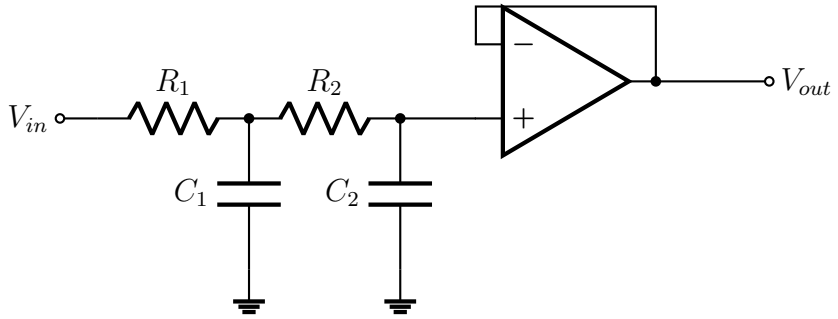
3.3 QPSK Modulator

The QPSK modulator utilizes a bit-splitter, two multipliers with a local oscillator, a 2-bit serial-to-parallel converter, and a summer circuit. At the input, the message signal's even bits (e.g., 2nd, 4th, 6th) and odd bits (e.g., 1st, 3rd, 5th) are separated by the bit-splitter. These are multiplied with the same carrier to generate odd BPSK (called PSK_I) and even BPSK (called PSK_Q). The carrier for the PSK_Q signal is phase-shifted by 90° before modulation.

4 Circuit Implementation

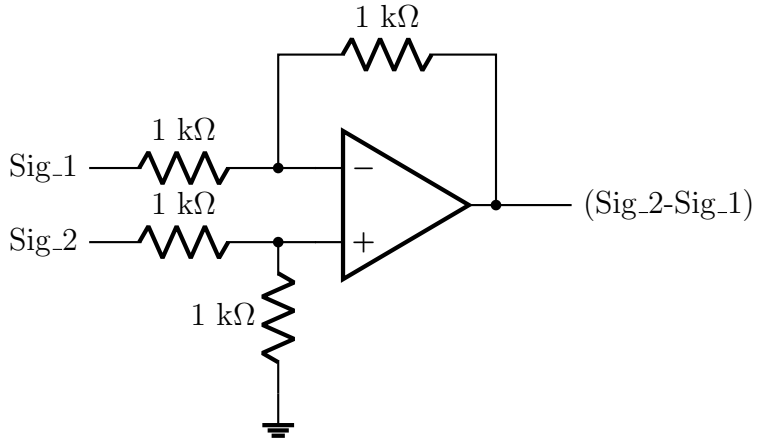
4.1 Phase Shifter (Generating Cosine from Sine)

An Op-Amp-based differentiator circuit is used to apply a 90° phase shift to the incoming sine-wave carrier, thereby generating a cosine wave.

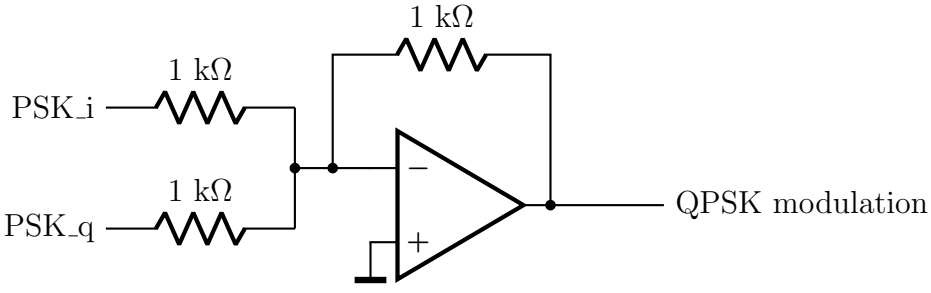


4.2 Subtractor and Adder Using Op-Amp

- **Subtractor:** An Op-Amp subtractor is utilized to generate inverted carrier signals (i.e., generating $-\sin$ from sin, or $-\cos$ from cos).

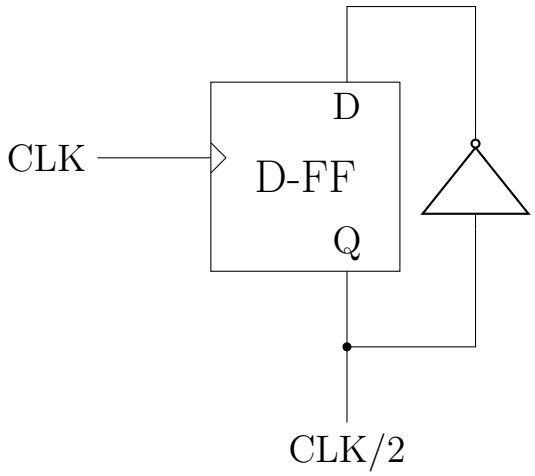


- **Adder:** An Op-Amp adder is used at the final stage to sum the PSK_I and PSK_Q signals, producing the final QPSK modulated output.



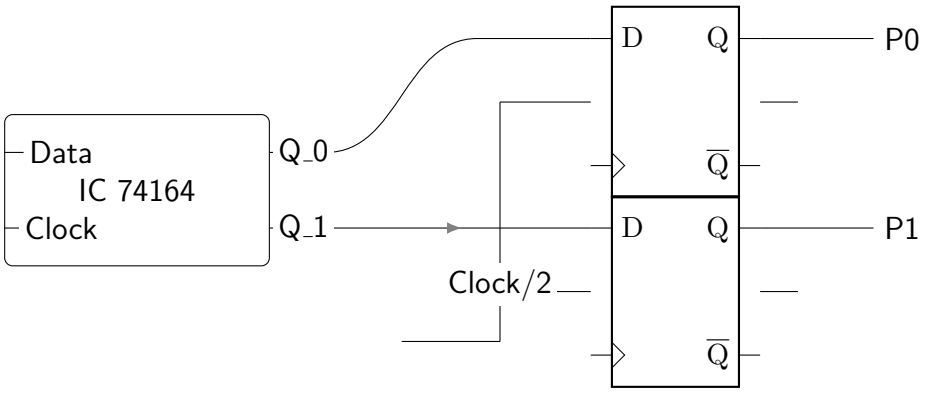
4.3 Clock Division (Clock/2) using D Flip-Flop and NOT Gate

A divide-by-2 clock is generated by configuring a D flip-flop in toggle mode. The output Q is inverted using a NOT gate and fed back to the D input, so the state toggles on every clock edge and produces a stable $f/2$ clock signal.



4.4 Serial to Parallel Data Implementation using Shift Register and D Flip-Flop

The incoming serial bit stream is converted into two synchronized parallel branches (odd and even bit streams) before QPSK modulation. This function is implemented using a 7-bit shift register together with a D flip-flop.



- The serial data sequence is first applied to the input of IC 74164 (shift register) and shifted on each active clock edge.
- A D flip-flop (IC 7474) is driven using a clock at half the shift-register clock frequency so that alternate bits are separated correctly.
- One branch carries odd-indexed bits and the other carries even-indexed bits, forming the I-channel and Q-channel data inputs.
- These two outputs are then fed to the switching/multiplier stages for generating PSK_I and PSK_Q prior to summation.

4.5 Complete QPSK Modulation Schematic

The full schematic integrates the bit-splitter (using a 7-bit shift register and D flip-flop), the phase shifters, the analog switches (IC M4016) acting as multipliers, and the final summing amplifier.

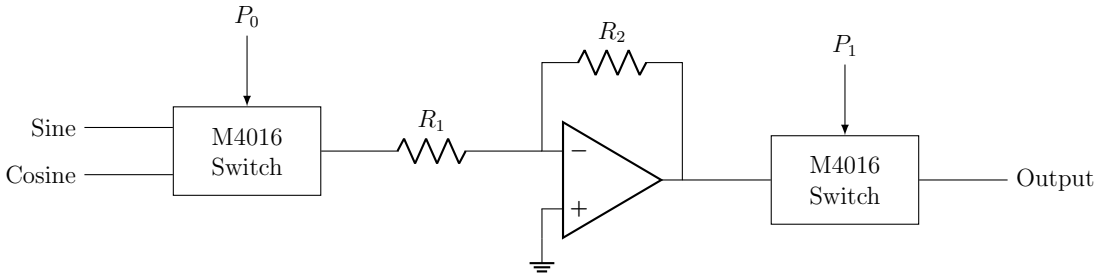


Fig: Modulation Circuit

5 Observations and Results

5.1 QPSK Modulation Hardware Implementation

The QPSK modulator was implemented using a combination of digital logic for data handling and analog circuitry for carrier generation and routing.

- **Serial-to-Parallel Conversion:** The input serial bit stream was processed using a 7-bit shift register (IC 74164). To extract 2-bit parallel data (dibits), the outputs Q_0 and Q_1 were read at half the frequency of the master write clock.
- **Clock Division:** To generate the required half-frequency clock ($f/2$), a D flip-flop (IC 7474) was utilized. By connecting the inverted output ($\bar{Q}$) back to the D input and applying the master clock to the CLK pin, a stable $f/2$ clock was successfully created to trigger the parallel data extraction.
- **Carrier Generation and Phase Shifting:** The primary carrier was a sine wave. A differentiator circuit was used to generate a 90° phase-shifted cosine wave. To obtain the four required phases ($+\sin, -\sin, +\cos, -\cos$), inverting amplifiers were implemented to generate the negative counterparts.
- **Constellation Pivot:** Initially, the design aimed to form a constellation where each symbol was represented by $a\sin(x) + b\cos(x)$ (where $a, b \in \{+1, -1\}$). However, this approach significantly increased circuit complexity and introduced excessive noise. To mitigate this, we switched to a simplified constellation where symbols are represented directly by $+\sin$, $-\sin$, $+\cos$, or $-\cos$ depending on the input dibit. This reduced the number of active components and produced an output that was less prone to noise.
- **Signal Routing:** An M4016 analog switch IC was used to route the specific carrier signals based on the parallel data extracted from the D flip-flop.

5.2 Measured Output Results

The following oscilloscope captures show the experimentally obtained outputs for the phase-shifter stage and the final QPSK modulation stage.

5.2.1 Cosine from Sine Output

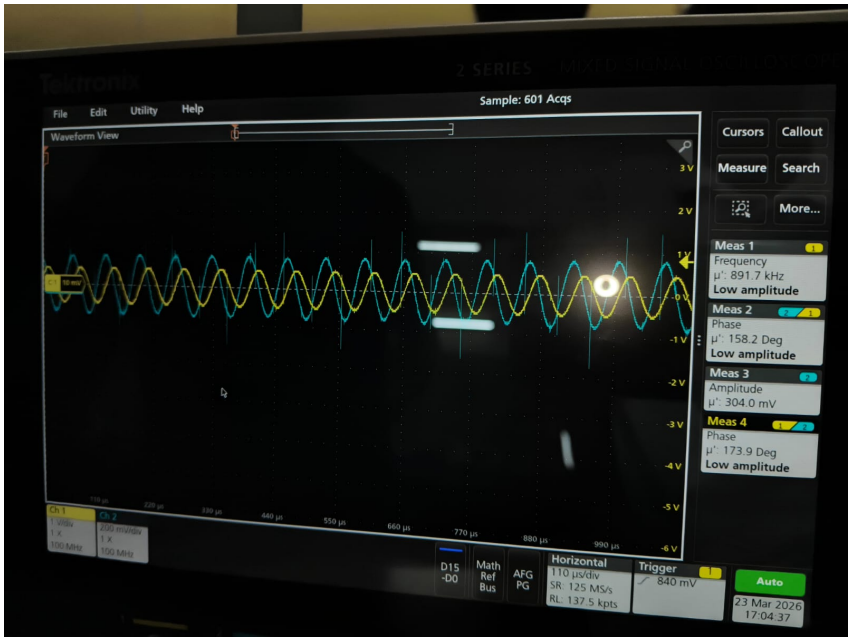


Figure 1: Measured output of the phase-shifter circuit showing cosine generation from an input sine signal.

5.2.2 QPSK Modulator Output

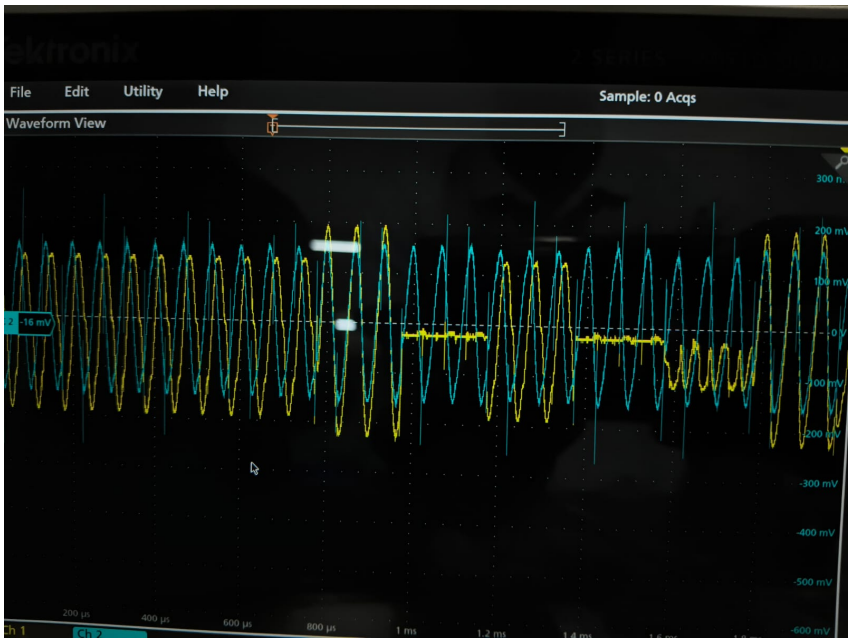


Figure 2: Measured QPSK modulator output waveform obtained from the hardware implementation.

5.3 QPSK Demodulation Approach

Due to the high noise levels present in the modulator's hardware output, real-time hardware demodulation could not be completed. However, the signal can be effectively demodulated offline using MATLAB if the data is captured in CSV format from the oscilloscope.

- **Data Acquisition:** The modulated signal and the reference time vector are imported from the CSV file using the `readtable` function.
- **Pre-processing:** The signal is first normalized, and any DC offset is removed to ensure the In-phase (I) and Quadrature (Q) components are centered.
- **Coherent Detection:** The received signal Rx_{sig} is multiplied by locally generated synchronized carriers:

$$I = Rx_{sig} \cdot \cos(2\pi f_c t) \quad \text{and} \quad Q = Rx_{sig} \cdot \sin(2\pi f_c t) \tag{1}$$

- **Noise Filtering and Downsampling:** The resulting I and Q signals are passed through a digital low-pass filter (LPF) to remove the high-frequency components at $2f_c$. The signal is then downsampled to reduce computational complexity.
- **Symbol Recovery:** By averaging the filtered I and Q values over each symbol period, a threshold decision (e.g., $I > 0$ or $Q > 0$) is applied to recover the original transmitted bit stream.

6 Discussion

- **Design Optimization:** The shift from a composite $a\sin(x) + b\cos(x)$ signal to a direct phase-switching model ($\pm\sin, \pm\cos$) was a critical engineering decision. It demonstrated that reducing circuit complexity is often more effective than complex summation for maintaining signal integrity in a noisy environment.
- **Hardware Constraints:** The noise observed in the oscilloscope waveforms likely originated from switching transients in the M4016 and non-idealities in the op-amp-based differentiator. These factors prevented hardware-based recovery but served as a practical lesson in the importance of Signal-to-Noise Ratio (SNR) in digital communication.
- **Bandwidth Efficiency:** The implementation of the $f/2$ clock for the shift register confirmed the theoretical advantage of QPSK: it reduces the symbol rate to half the bit rate ($R_s = R_b/2$), thereby doubling the bandwidth efficiency compared to BPSK.
- **Synchronization Issues:** FFT analysis showed the carrier frequency shifting from 2 kHz to 1.7 kHz. This discrepancy emphasizes the necessity of accurate carrier synchronization and suggests that sampling frequency mismatches or limited FFT resolution can significantly impact demodulation accuracy.